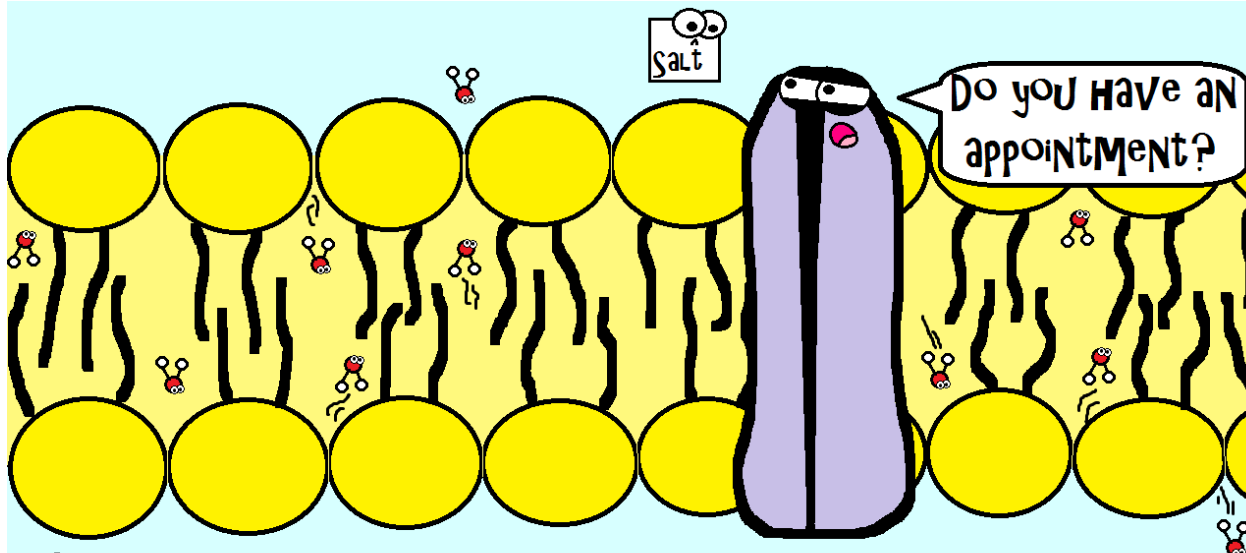


Deciphering the molecular basis and pharmacotherapy of Cystic Fibrosis

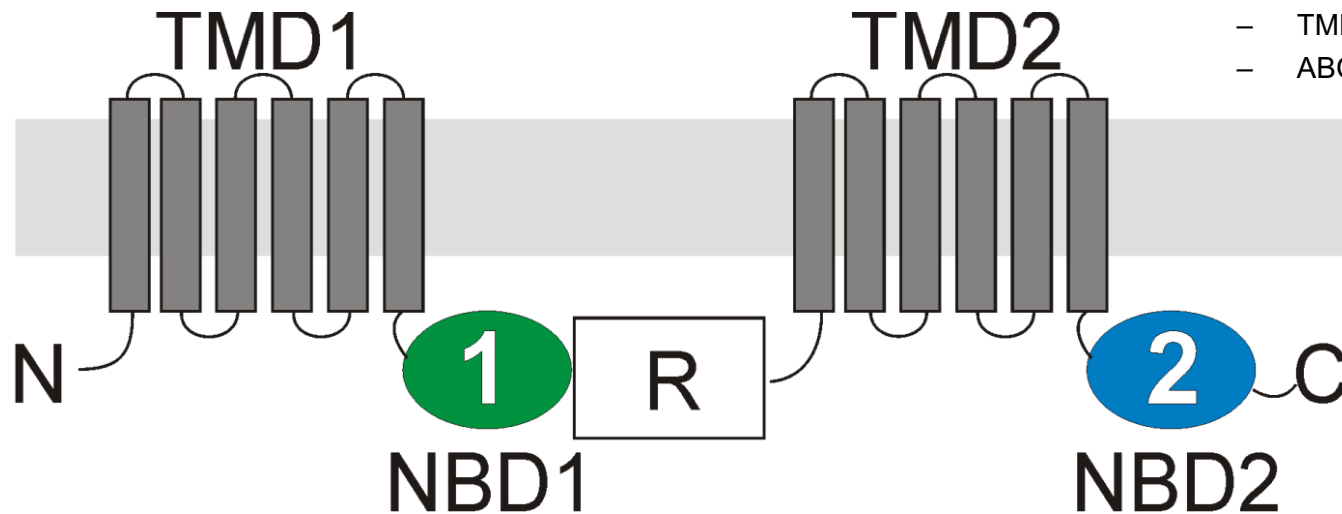


Zheheng Fan

22009451

Biomedical Sciences *BSc.*

Supervisor: Dr. Paula Vergani

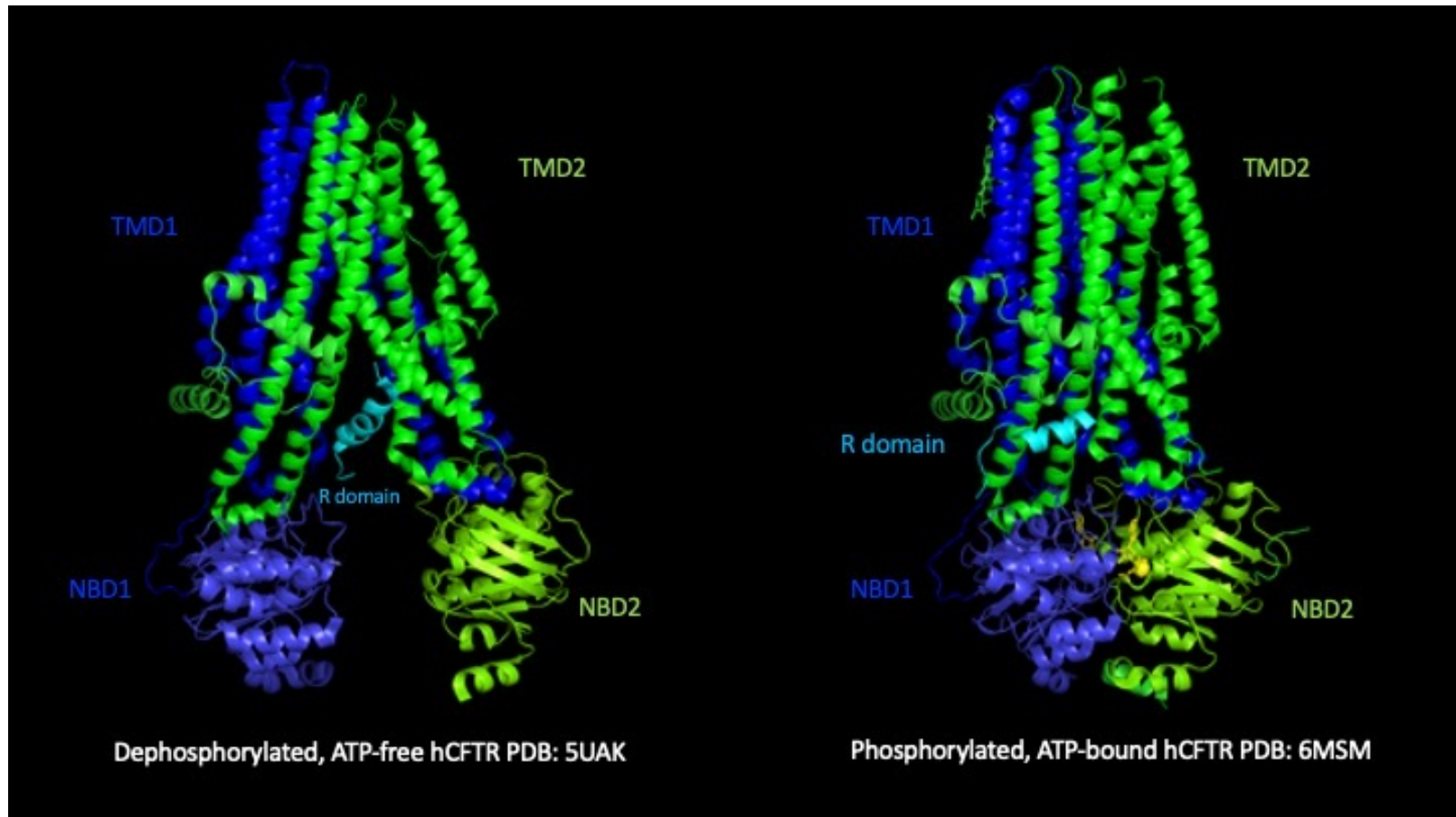


- **Glossary**

- NBD – Nucleotide-binding domain
- TMD – Transmembrane domain
- ABC – ATP-binding cassette

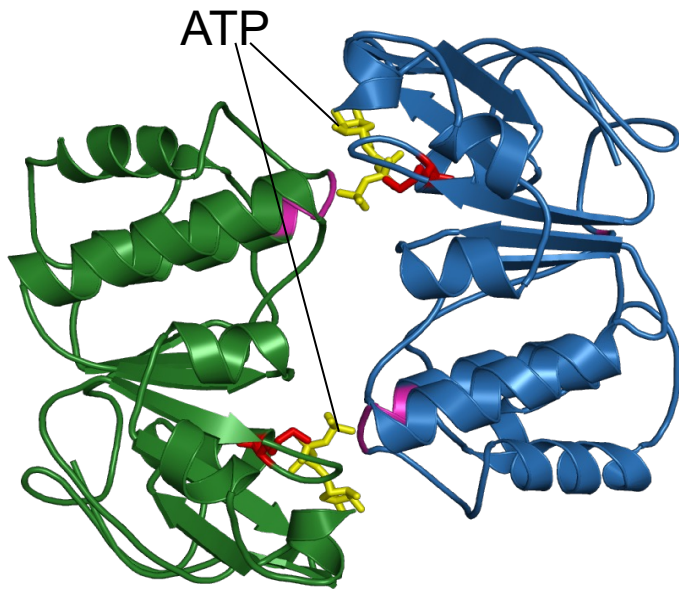
- CFTR is a unique member of the ABC transporter family
- Main function:
 - Allow the flow of Cl^- and HCO_3^- passively down the concentration gradient
- 5 domains:
 - two membrane-spanning TMDs, two NBDs, one regulatory R domain

R-domain phosphorylation regulates CFTR channel opening

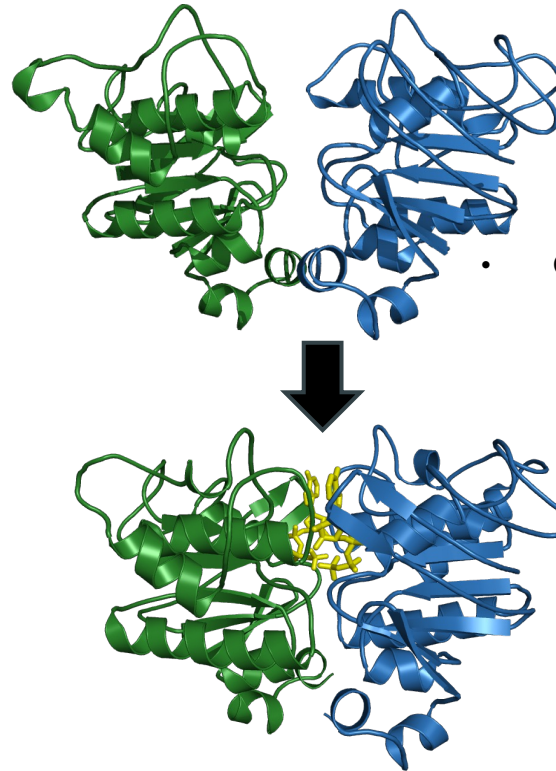
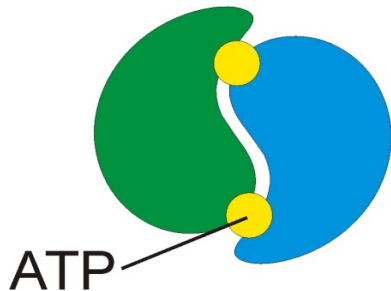


- R domain is phosphorylated by cAMP-dependent kinase A (PKA) and protein kinase C (PKC)
- Phosphorylation of the R domain is a prerequisite for channel opening

ATP-dependent dimerization of NBDs drives CFTR channel opening



MalK (*E.coli*)
Chen *et al.*, 2003



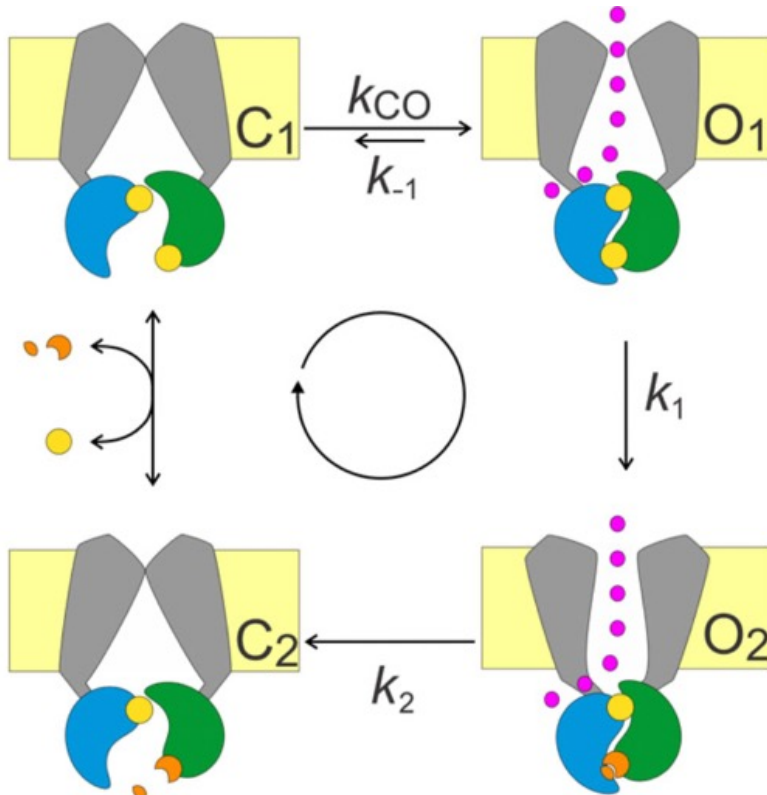
Glossary

- Signature motif/ Lasso motif/ LSGGQ motif → Labelled pink
- Walker A and B motif → Labelled red

ATP-bound dimerized NBDs

- Two NBDs form a rotationally symmetrical head-to-tail dimer
- The dimerization signal is transduced to the TMDs to allow channel opening

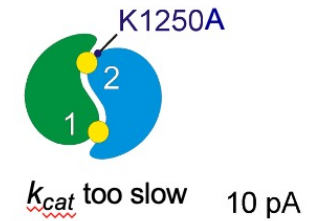
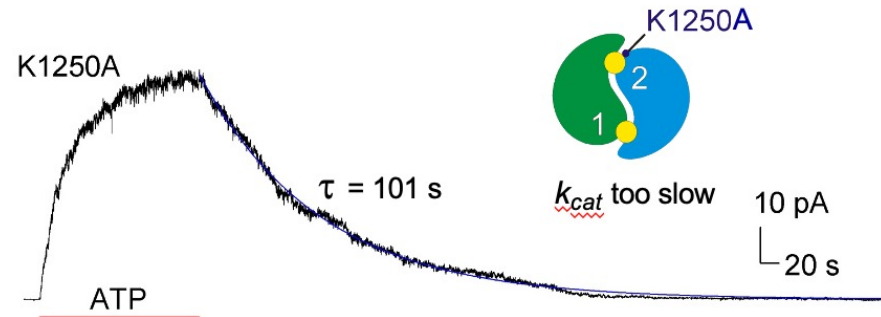
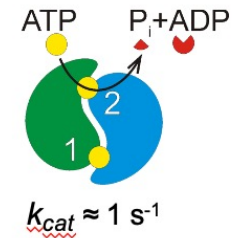
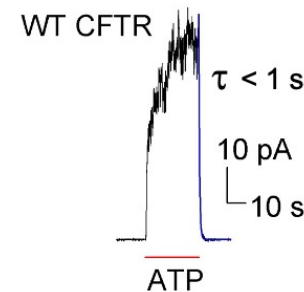
ATP hydrolysis and gating cycles



Langron *et al.* (2018)

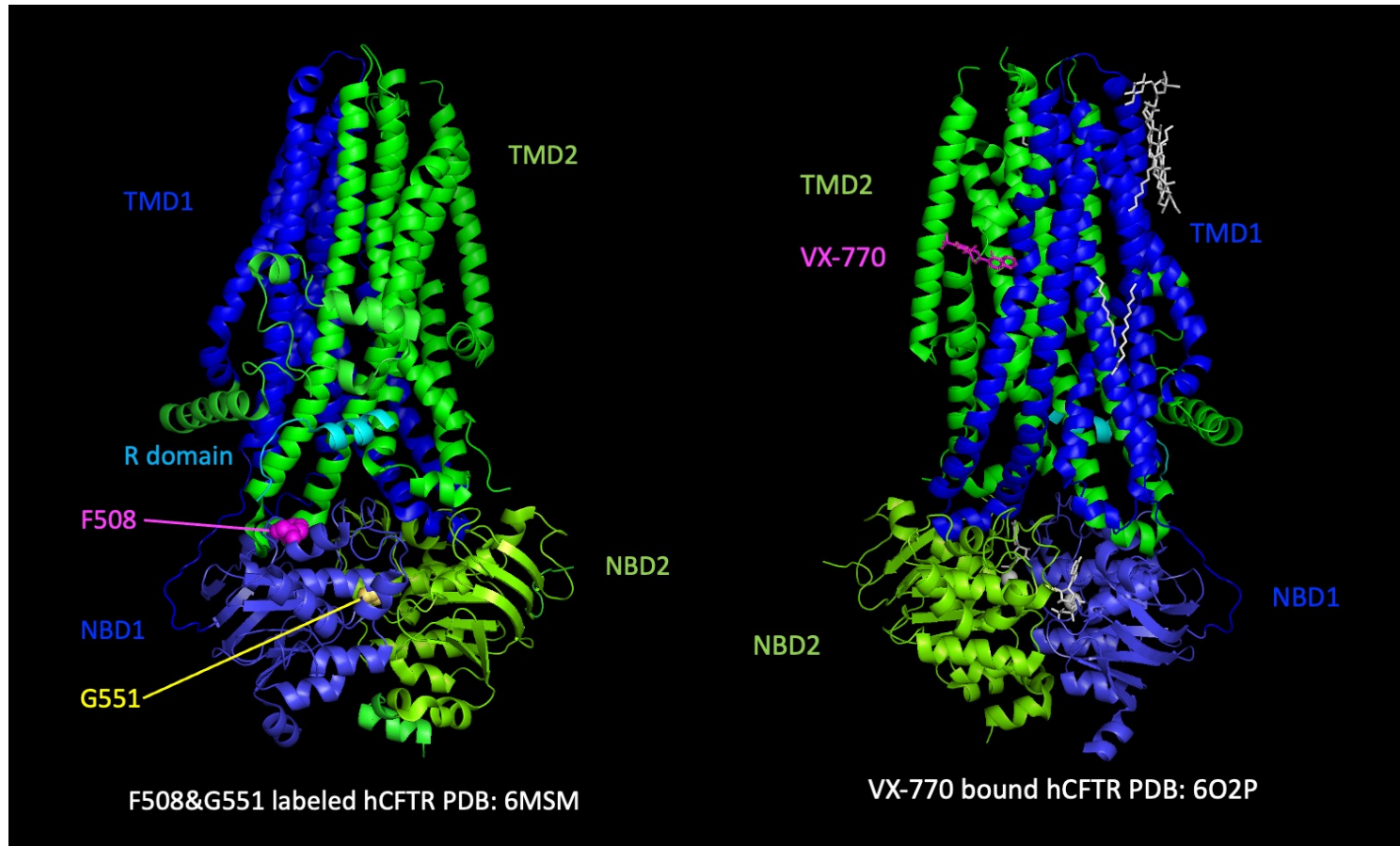
- Most WT-CFTR favours the hydrolytic pathway (C₁ → O₁ → O₂ → C₂ → C₁)
- For mutants that abolish ATP hydrolysis at site 2, they undergo the non-hydrolytic pathway (C₁ → O₁ → C₁)

- **Glossary**
 - K1250 – Walker A lysine in NBD2, forming composite site 2
 - τ_b -- burst duration
 - K1250A is not a disease-causing mutation



- K1250A mutation greatly increases the burst duration and it closes completely via the non-hydrolytic pathway

Common CFTR mutations and CFTR potentiator

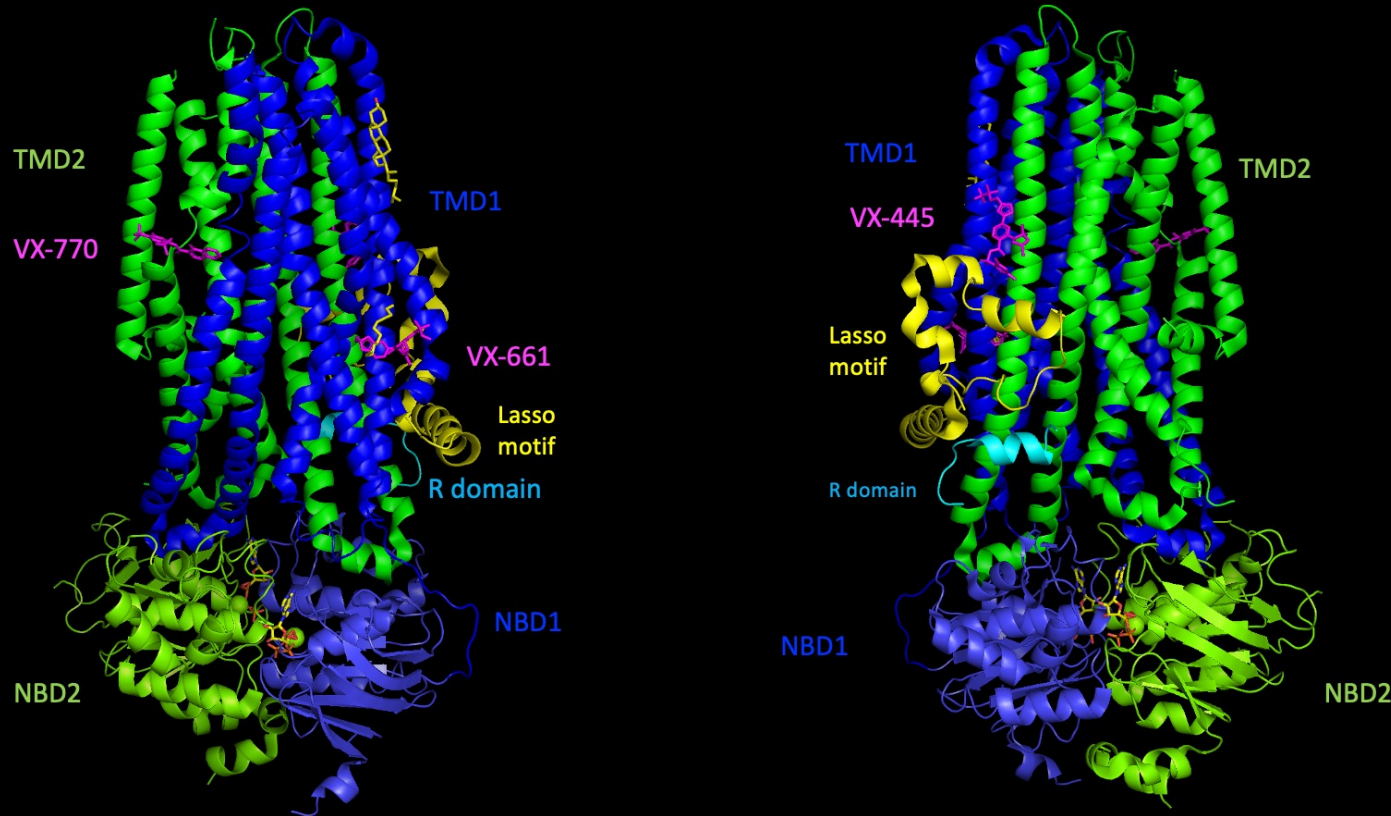


Glossary

VX770 (Ivacaftor)
-- First FDA-approved
CFTR potentiator for
G551D mutation

- VX770 increases the open probability of CFTR by stabilizing the pre-hydrolytic O_1 state
- VX770 is life-changing for CFTR gating defect mutations → combination with correctors necessary for rescuing defective folding mutations

CFTR correctors in combination with potentiators



Two views of Trikafta (VX-770+VX-661+VX-445) bound hCFTR PDB: 8EIQ

Glossary

Trikafta – Approved by FDA for patients with one copy of F508del

VX-661 (Tezacaftor) – Type I corrector

VX-445 (Elexacaftor) – Type III corrector

VX-770 (Ivacaftor) – Potentiator

- Trikafta = VX-770 + VX-661 + VX-445
- VX-661 and VX-445 act synergistically and greatly improve the biogenesis of CFTR
- Trikafta greatly improves lung function of F508del patients

Thank you!

Q&A

